12 Prove that the square of an odd number is always 1 more than a multiple of 4

Let the odd number be $2n + 1$, where n is an integer.

1 mark
Represent every odd integer

×	$2n$	1
$2n$	$4n^2$	$2n$
1	$2n$	1

1 mark
Expand $(2n + 1)(2n + 1)$
using every grid product

$(2n + 1)^2 = 4n^2 + 2n + 2n + 1$

$= 4n^2 + 4n + 1 = 4(n^2 + n) + 1$

1 mark
Correct expansion

$n^2 + n$ is an integer, so $4(n^2 + n)$ is a multiple of 4.

Therefore the square is exactly 1 more than a multiple of 4.

1 mark
Use the algebra to justify the statement

(Total for Question 12 is 4 marks)

13 $\sqrt{5}(\sqrt{8} + \sqrt{18})$ can be written in the form $a\sqrt{10}$ where a is an integer.

Find the value of a.

$a =$ .....

(Total for Question 13 is 3 marks)

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